ARCUS-H: Behavioral Stability Evaluation Under Controlled Stress as a Complementary Axis for Reinforcement Learning Assessment

Karim ZINEBI
Independent Researcher
karim.zinebi@nuraql.com

Abstract

Episodic return measures task performance but is insufficient as a sole evaluation criterion: policies that appear strong under nominal conditions may degrade substantially when sensors are noisy, actuators are unreliable, or reward feedback is corrupted. We present ARCUS-H (*Adaptive Reinforcement Coherence Under Stress—Harness*), a post-hoc evaluation harness that applies structured stress protocols to any trained Stable-Baselines3 policy—without retraining or model-internal access—and measures behavioral stability through five formally-defined interpretable channels: Competence, Policy Consistency, Temporal Stability, Observation Reliability, and Action Entropy Divergence, each grounded in the MDP trajectory structure and information-theoretic principles.

We evaluate **51 (environment, algorithm) pairs** across **12 environments**, **8 algorithms**, **8 stress schedules**, and **10 seeds**, totalling **979,200 evaluation episodes**. The primary correlation between ARCUS stability scores and normalized reward—computed on env-axis stressors only to avoid reward-corruption circularity—is $r = +0.240$ (95% CI: $[0.111, 0.354]$, $p = 1.12 \times 10^{-4}$, $n = 255$ policy-level), implying that **94.3%** of stability variance is unexplained by return ($R^2 = 0.057$). CVaR-5 and ARCUS are positively correlated ($r = +0.355$, $p < 10^{-8}$) yet not interchangeable: 10.6% of policies show high CVaR tail risk but low ARCUS degeneracy—cases where return variance is high but behavioral structure remains stable under stress. Three deployment-relevant empirical findings emerge: (1) SAC collapses at 92.5% under observation noise versus 61.0% for TD3 under identical conditions, a 31.5pp difference explained by their contrasting entropy objectives; (2) MuJoCo policies degrade at 78.6% on average despite achieving the highest returns of any suite tested; and (3) SpaceInvaders CNN policies collapse at only 13.0% under noise that degrades Pong at 41.9%, demonstrating that robustness is representation-dependent, not architecture-determined.

1 Introduction

Standard RL benchmarks report expected episodic return under ideal, stationary conditions. Return is a necessary signal—it tells us whether a policy achieves its objective in training conditions. But it is not sufficient for deployment assessment. It assumes faithful observations, reliable action execution, and stable reward feedback—assumptions that hold in simulation but fail routinely in deployment. Under sensor noise, actuator wear, covariate shift, or reward corruption, two agents with identical leaderboard return can exhibit radically different behavioral profiles when conditions deviate from training.

Why existing metrics are insufficient. Return variance and CVaR [9, 8] capture tail risk on the *return distribution* but say nothing about *how* behavior changes under stress—whether actions become erratic, whether the behavioral signature drifts from a pre-deployment anchor, or whether the policy recovers after stress ends. ARCUS-H operates on *behavioral channel trajectories* rather than scalar returns, providing complementary diagnostic information. Adversarially robust RL [6, 7] requires modified training objectives; ARCUS-H is evaluation-only and requires no training modification. Distributional RL [10] captures the full return distribution but not behavioral stability channels. RL reliability metrics [3] address statistical reproducibility across seeds, not stress-induced behavioral degradation.

Formal grounding. Each ARCUS-H channel is derived directly from MDP trajectory statistics without requiring model internals (Section 3.2). The entropy proxy (Meaning channel) uses PCA whitening and per-principal-component histogram entropy, grounded in standard information-theoretic arguments about marginal vs. joint entropy in high-dimensional spaces. The adaptive threshold calibration is a standard non-parametric quantile estimator applied to the pre-phase score distribution, achieving FPR$\approx \alpha$ by construction.

On the primary correlation value. $r = +0.240$ represents a weak-to-moderate positive relationship. The important quantity is $R^2 = 0.057$: return explains 5.7% of stability variance. ARCUS-H measures the other 94.3%. A positive correlation is expected and necessary—a random policy achieves neither high return nor high stability—but its weakness is the empirical justification for ARCUS-H as a complementary axis.

Novelty over existing frameworks. ARCUS-H differs from generalization benchmarks (ALE [14], Procgen [13]) by applying *post-hoc* structured stress to *already-trained* policies—not variants of the training environment. It differs from robust RL by requiring no training modification. It differs from reliability

metrics [3] by measuring behavioral channel degradation, not seed variance of return. The closest related work, CARL [17], varies environment parameters but provides no behavioral channel decomposition, no multi-phase protocol, and no adaptive calibration.

Contributions.

1. A formally-defined post-hoc behavioral stability harness: three-phase pre/shock/post protocol, eight stressors spanning three orthogonal failure axes, no retraining, no model-internal access.
2. Five behavioral channels derived from MDP trajectory structure, with adaptive inverse-MAD weighting that is principled (channels with low baseline variance are less informative detectors) and validated against per-channel MAD data.
3. A PCA-whitened joint entropy proxy for the Meaning channel, extending information-theoretic measurement to high-dimensional continuous action spaces.
4. The largest-scale empirical evaluation of RL behavioral robustness to date: 979,200 evaluation episodes across 51 policy configurations.
5. Three deployment-relevant findings with mechanistic explanations: SAC/TD3 architectural fragility, MuJoCo structural vulnerability, and CNN representation-dependent robustness.
6. Quantified CVaR vs. ARCUS complementarity: correlated but not substitutable.

2 Related Work

Reproducibility and evaluation reliability. Henderson et al. [1] show that RL results are highly sensitive to random seeds and hyperparameters. Agarwal et al. [2] propose stratified bootstrap confidence intervals. Chan et al. [3] introduce reliability metrics beyond mean return. These works address the question of whether return *comparisons* are statistically reliable; ARCUS-H asks whether return is the *right* metric for deployment assessment.

Risk-sensitive RL. CVaR [9] and risk-sensitive MDPs [8] optimize tail return during training. Distributional RL [10] models the full return distribution. Section 6.3 provides the first empirical quantification of CVaR vs. ARCUS complementarity: they are correlated ($r = +0.355$) but measure distinct dimensions.

Robust RL. Minimax approaches [5], adversarial training [6], and action-robust RL [7] incorporate stress into training objectives. ARCUS-H is evaluation-only.

Evaluation frameworks. ALE [14] and Procgen [13] benchmark generalization. DMControl [15] targets continuous control. CARL [17] varies context parameters. None provides post-hoc structured stress evaluation with behavioral channel decomposition and adaptive calibration.

3 ARCUS-H Framework

3.1 MDP Setup

Let $M = (S, A, P, r, \gamma, T)$ be a finite-horizon MDP and $\pi_\theta : S \to \Delta(A)$ a pretrained policy. An evaluation episode produces trajectory $\tau = (s_1, a_1, \ldots, s_T, a_T)$ with return $G(\tau) = \sum_{t=1}^{T} \gamma^{t-1} r(s_t, a_t)$.

3.2 Behavioral Channels

Five per-episode channels $c_e, h_e, t_e, i_e, m_e \in [0, 1]$ (higher = more stable). All are derived from observable trajectory statistics; none require internal model access.

Competence (c_e).

$$c_e = \sigma\left(6 \cdot \frac{R_e - \bar{R}_{e-1}}{S_R}\right), \qquad \sigma(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

where $\bar{R}_{e-1}$ is the exponential moving average of returns and S_R its EMA scale. The logistic maps relative return improvement to $(0, 1)$; the factor 6 gives σ approximately 0.05 and 0.95 at ± 1 standard unit.

Policy Consistency (h_e, Coherence). For discrete actions: $h_e = 1 - \rho_e$, where $\rho_e \in [0, 1]$ is the within-episode action-switch rate. For continuous actions:

$$h_e = \exp\left(-2 \max\left(0, \frac{1}{T_e - 2} \sum_{t=3}^{T_e} \|\mathbf{j}_{e,t}\|_2\right)\right) \tag{2}$$

where $\mathbf{j}_{e,t} = (\mathbf{a}_{e,t} - \mathbf{a}_{e,t-1}) - (\mathbf{a}_{e,t-1} - \mathbf{a}_{e,t-2})$ is the second-order action finite difference (jerk). Eq. (2) penalises erratic action sequences; the exponential ensures $h_e \in (0, 1]$.

Note on negative drops: Under resource constraint and sensor blackout, some agents reduce action variability, causing coherence to *improve*. The radar chart (Fig. 6) clips such negative drops to zero and annotates them. This is a feature, not a failure—it reveals that those stressors can induce over-conservative behavior.

Temporal Stability (t_e, Continuity).

$$t_e = \exp\left(-\frac{\|\mathbf{s}_e - \mathbf{s}_{e-1}\|_2}{\sigma_c}\right) \tag{3}$$

where $\mathbf{s}_e$ is a compact behavior-signature vector of action and reward statistics for episode e. Measures episode-to-episode behavioral self-consistency.

Observation Reliability (i_e, Integrity).

$$i_e = \exp\left(-\frac{\|\mathbf{s}_e - \mathbf{a}\|_2}{\sigma_i}\right) \tag{4}$$

where $\mathbf{a}$ is the running-mean pre-phase behavioral anchor updated during the first 40 episodes. Measures fidelity to the nominal behavioral fingerprint.

Action Entropy Divergence (m_e, Meaning).

$$m_e = \exp\big(-\alpha_{\text{eff}} \max(0, H_e - \bar{H}_{\text{pre}})\big) \tag{5}$$

where H_e is the joint entropy of the episode's action distribution, approximated via PCA whitening. Specifically, we apply SVD to the $T_e \times D$ action matrix to obtain principal-component projections, compute per-PC histogram entropy H_k using a fixed number of bins, and set $H_e = \frac{1}{D}\sum_k H_k$. The PCA step corrects the marginal-independence assumption that would underestimate entropy in high-dimensional spaces (e.g., Walker2d's 6-DoF actions). m_e is high when entropy diverges upward from the pre-phase baseline, indicating loss of goal-directed action structure.

Entropy estimation scope: The histogram-based PC entropy captures linear correlations but not non-Gaussian inter-PC coupling. For $D > 6$, copula-based or kernel-density estimators would be more rigorous; this is noted in Section 8.

Why five channels, not one? A single composite score would conflate mechanistically distinct failure modes. The five channels correspond to five diagnostic questions: (Competence) is the policy performing worse than its recent baseline? (Consistency) are actions becoming more erratic? (Continuity) is episode-to-episode behavior changing rapidly? (Integrity) is behavior diverging from the nominal fingerprint? (Meaning) is action structure becoming less goal-directed? Fig. 14 demonstrates empirically that the five channels are non-redundant: each stressor produces a distinct density pattern per channel.

3.3 Collapse Scoring and Adaptive Threshold

Discriminative MAD weighting. Pre-phase channel variances are estimated via the median absolute deviation (MAD):

$$w_k = \frac{\sqrt{\text{MAD}_k}}{\sum_j \sqrt{\text{MAD}_j} + \varepsilon} \tag{6}$$

Channels with low pre-phase MAD contribute less to the collapse score because near-constant channels provide less discriminative power. Square-root damping prevents high-variance channels from dominating. Fig. 12 validates that MAD weights reflect genuine per-channel signal: shock-phase MAD is consistently higher than pre-phase MAD, confirming that channels do respond to stressors.

Sensitivity analysis: Rankings were qualitatively stable under $\pm 20\%$ weight perturbations across all 51 policy configurations. The MAD weighting is a principled heuristic; alternative approaches (e.g., information-gain weighting, learned attention) are future work.

Collapse score. From pre-phase episodes, ARCUS-H computes robust z-scores for each channel. The raw collapse score is the weighted Euclidean norm in z-score space:

$$\text{raw}_e = \sqrt{\sum_k w_k z_{k,e}^2} \tag{7}$$

This is a Mahalanobis-style distance from the pre-phase center. The final collapse score $S_e \in [0, 1]$ is obtained by passing raw_e through a logistic sigmoid with sharpness β and center γ.

Adaptive threshold.

$$\eta = \text{p}_{95}(\{S_e : e \in \mathcal{T}_{\text{pre}}\}) \tag{8}$$

This is a standard non-parametric quantile estimator applied to the pre-phase score distribution. By construction, $\Pr[S_e > \eta \mid e \in \mathcal{T}_{\text{pre}}] \approx 0.05$, achieving FPR$\approx \alpha = 0.05$ without any environment-specific tuning. The p95 threshold is recomputed independently for each (env, algo, seed) run.

3.4 Leaderboard Score

$$\mathcal{L} = 0.55\,\bar{I} + 0.30\,(1 - CR_{\text{shock}}) + 0.15\,r_{\text{norm}} \tag{9}$$

where $\bar{I}$ is the mean identity score (composite of all five channels), CR_{shock} is the shock-phase collapse rate, and $r_{\text{norm}} \in [0, 1]$ is per-environment min-max normalized reward. The 15% reward component is intentional and declared: a stable but ineffective policy should not rank equally with a stable and effective one. Its presence is precisely why VI and RN are excluded from the primary correlation analysis—those stressors corrupt the reward signal, creating circularity through r_{norm} in Eq. (9).

4 Stress Schedules

During the shock phase, the harness intercepts each environment step and applies one stressor. Let $a_t \sim \pi_\theta(\cdot|s_t)$, s_t be the nominal observation, r_t the nominal reward.

Perception axis (3 stressors). Observation Noise (ON). $s_t^{\text{exec}} = s_t + \boldsymbol{\epsilon}_t$, $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{ON}}^2 \mathbf{I})$, $\sigma_{\text{ON}} = 0.15\hat{\sigma}_s$. Models i.i.d. sensor noise and quantization error. Each step is perturbed independently; there is no temporal correlation.

Sensor Blackout (SB). With probability p_{SB} per step, a contiguous window of $[N_{\min}, N_{\max}]$ steps receives $s_t^{\text{exec}} = \mathbf{0}$. Unlike ON, SB has temporal correlation (contiguous windows). Models camera dropout, GPS blackout, and packet loss. The channel signature difference between ON and SB (Fig. 7) confirms they are non-redundant despite attacking the same axis.

Concept Drift (CD). $s_t^{\text{exec}} = s_t + \boldsymbol{\delta}_t$, $\boldsymbol{\delta}_t = \boldsymbol{\delta}_{t-1} + \boldsymbol{\epsilon}_t$ (cumulative drift). Models covariate shift in deployment and sim-to-real mismatch. Key property: the perturbation is non-stationary and grows over the shock phase.

Execution axis (2 stressors). **Trust Violation (TV).** Continuous: $a_t^{\text{exec}} = Ma_t + \boldsymbol{\epsilon}_t$, where M is a fixed diagonal distortion matrix. Discrete: $a_t^{\text{exec}} = \pi_f(a_t)$ under a fixed non-identity permutation. Models actuator wear, miscalibration, and action-execution mismatch.

Resource Constraint (RC). Continuous: $a_t^{\text{exec}} = \kappa a_t$, $\kappa \in (0, 1]$. Discrete: default action with probability p. Models reduced control authority, rate limiting, and communication bandwidth constraints.

Feedback axis (2 stressors, excluded from primary correlation). **Valence Inversion (VI).** $r_t^{\text{exec}} = -r(s_t, a_t^{\text{exec}})$. Reward sign inversion. Excluded from the primary correlation analysis because it corrupts the reward signal, which enters Eq. (9) directly (circularity). Included as an upper-bound severity stressor.

Reward Noise (RN). $r_t^{\text{exec}} = r_t + \nu$, $\nu \sim \mathcal{N}(0, \sigma_{\text{RN}}^2)$. Stochastic reward corruption. Also excluded from primary correlation for the same reason.

Stressor calibration and realism. All stressors are calibrated to 15% of observed signal statistics (standard deviation of observations, reward scale) from a reference pass on each environment. This is conservative by design—the perturbation is commensurate with natural observation variability. ON matches sensor noise intensities documented in robotics benchmarks [18]; SB matches packet-loss models in UAV communication [19]; CD matches covariate shift in sim-to-real transfer [20]; TV matches actuator wear models [7]; RC matches sparse-reward and rate-limited deployments. The stressors are not arbitrary—each corresponds to a documented failure mode in deployed RL systems.

5 Experimental Protocol

Environments. 12 environments spanning four suites:

- **Classic control** (5): CartPole-v1, Acrobot-v1, FrozenLake-v1, MountainCar-v0, LunarLander-v3
- **Continuous control** (2): MountainCarContinuous-v0, Pendulum-v1
- **MuJoCo locomotion** (3): HalfCheetah-v4, Hopper-v4, <u>Walker2d-v4</u>
- **Atari CNN** (2): ALE/Pong-v5, <u>ALE/SpaceInvaders-v5</u>

Underlined = new in this evaluation.

Algorithms. 8 algorithms via SB3 [16] and SB3-Contrib: On-policy: PPO, A2C, TRPO; Value-based: DQN, QR-DQN; Off-policy AC: DDPG, SAC, TD3. Not all algorithms are applicable to all environments (e.g., DQN requires discrete actions).

Scale. 51 (env, algo) pairs; 10 seeds (0–9); 120 episodes per run (40/40/40 pre/shock/post); deterministic and stochastic eval modes; 979,200 evaluation episodes in total.

Training budgets. Classic control: 3×10^5 steps. Continuous: 3×10^5 steps. MuJoCo: 1×10^6 steps. Atari: 3×10^6 steps (PPO, A2C). Walker2d (PPO, A2C): 3×10^6 steps. Empirical FPR for Walker2d $= 0.053$ (target 0.05), confirming convergence.

Limitation: Policies are trained with standard budgets, not tuned for maximum performance. The fragility findings reflect competently-trained policies, not state-of-the-art. Highly-optimized policies may show different fragility profiles. See Section 8.

Atari wrapper symmetry. Both Atari games use identical AtariPreprocessing + FrameStack(4, 84×84) wrappers with `obs_normalize` enabled, making cross-game comparisons valid.

Reproducibility. Full configuration tables, per-run baseline statistics, and hyperparameters are provided in the code repository. Each run records seed, algorithm, environment, FPR target, observed FPR, and evaluation mode. All results are reproducible from the archived data at the Zenodo DOI below.

6 Results

6.1 FPR Validation

The adaptive p95 threshold achieves mean FPR$= 0.061$ across all (env, algo, seed, eval_mode) runs (target $\alpha = 0.05$, Fig. 10). The mild inflation above 0.05 reflects genuine pre-phase behavioral variance—the threshold is conservative, not miscalibrated. The right panel of Fig. 10 confirms that shock-phase degeneracy rates substantially exceed pre-phase rates for all stressors, validating discriminative power.

6.2 Reward as an Incomplete Proxy for Stability

Primary correlation. Fig. 2 shows ARCUS stability score vs. normalized reward across all 255 policy-level observations under env-axis stressors:

$$r = +0.240 \ [0.111,\ 0.354],\ p = 1.12\times10^{-4},\ n = 255$$

$R^2 = 0.057$: return explains 5.7% of stability variance. The remaining 94.3% is not captured by return. Table 2 reports all robustness variants.

Why circularity is avoided. VI and RN are excluded because they corrupt the reward signal, which enters $\mathcal{L}$ (Eq. 9) via r_{norm}. Including them raises r to $+0.311$ (Table 2)—an artefact of the circularity, not additional signal. The secondary value is reported for transparency, not as a claim.

Robustness checks. Spearman $\rho = +0.180$ ($p < 0.005$); per-env z-normed $r = +0.234$; non-Atari $r = +0.220$; Atari-only $r = +0.716$ (based on $n = 20$ Atari policy observations—the high value reflects the limited Atari diversity). All variants confirm the primary conclusion.

Rank disagreement. 89% of policies shift more than 10 rank positions between reward and ARCUS stability rankings (mean displacement: 75.0 positions, max: 237). Six agents are fragile (top-quartile reward, top-quartile degeneracy); 42 are robust (below-median reward, bottom-quartile degeneracy). These populations are invisible to return-only evaluation (Fig. 3).

6.3 CVaR-5 vs. ARCUS Stability

CVaR-5 and ARCUS are positively correlated ($r = +0.355$ vs. ARCUS score, $p = 5.7 \times 10^{-9}$; $r = -0.328$ vs. degeneracy rate, $p = 8.0 \times 10^{-8}$; Table 5), confirming they are not orthogonal. However, 27/255 policy observations (10.6%) show high CVaR-5 but low ARCUS degeneracy (Fig. 4)—policies where the return distribution has wide tails but behavioral channels remain stable under stress.

Why CVaR does not substitute for ARCUS. CVaR operates on the *return distribution* and captures which episodes have the worst returns. ARCUS operates on *behavioral channel trajectories* and captures whether action patterns, behavioral continuity, and goal-directed structure degrade under stress. A policy may have a wide return distribution (high CVaR) while maintaining stable channels—it may be operating in a stochastic environment where returns vary but the policy always acts sensibly. Conversely, a policy may have low CVaR (consistent returns) while exhibiting severe behavioral collapse under stress if the stress does not immediately affect return. The 10.6% disagreement rate represents policies that would be ranked differently by the two metrics—a practically significant discrepancy for deployment decisions.

6.4 SAC Entropy Objective Amplifies Sensor Fragility

SAC collapses at 92.5% under observation noise; TD3 collapses at 61.0% under the identical stressor on the same continuous-action environments, with the same training budget, both off-policy actor-critic (Fig. 5).

Mechanistic explanation. SAC's entropy-regularisation objective [11] maximises $\mathbb{E}[\sum_t r_t + \alpha H(\pi(\cdot|s_t))]$. Under observation noise, each corrupted observation $\tilde{s}_t = s_t + \epsilon_t$ induces a high-entropy policy response because the policy is specifically trained to maintain action diversity. This amplifies sensitivity to noise: a more deterministic policy would respond to noise with smaller action deviations. TD3's target policy smoothing [12] adds noise during *training* target computation ($\tilde{a} = a + \epsilon$, $\epsilon \sim \mathcal{N}(0, \tilde{\sigma})$), which desensitises the learned value function to small observation perturbations during evaluation.

Replication. First observed in a pilot evaluation on 47 pairs (SAC: 90.2%, TD3: 61.1%). Replicated here across 51 pairs and 10 seeds. The 31.5pp difference is consistent across all continuous-action environments and is explained by a known architectural property, not a measurement artefact.

Practical implication. For sensor-noisy deployments, TD3 is preferable over SAC even when SAC achieves higher return in simulation. Return alone cannot surface this: both algorithms may match on the simulation leaderboard while differing by 31.5pp under realistic sensor conditions.

6.5 MuJoCo Fragility Across Three Environments

MuJoCo policies degenerate at 78.6% on average across env stressors (Fig. 9), substantially higher than Classic control (47.8%) or Continuous control (47.9%). Walker2d-v4 (PPO + A2C, FPR$= 0.053$) collapses at 59.4%—lower than HalfCheetah and Hopper because on-policy methods are inherently less fragile than off-policy AC under these stressors. Resource constraint is the worst stressor for Walker2d (85.9%), consistent with reward compression destroying the on-policy gradient signal.

The MuJoCo fragility claim now rests on three locomotion environments with fundamentally different dynamics: HalfCheetah (planar, cheetah morphology, no balancing requirement), Hopper (1D, strict balancing), and Walker2d (2D, bipedal balance). The pattern holds across all three, confirming it reflects a property of high-dimensional continuous control under stress rather than an idiosyncrasy of a single environment.

Why MuJoCo is most fragile despite highest return. High-performance locomotion policies exploit precise continuous action dynamics—small deviations in action magnitude or direction cause large trajectory divergences due to the chain-reaction nature of rigid-body physics. This is structurally different from classic control (CartPole, Pendulum) where the dynamics are simpler and more forgiving.

6.6 CNN Robustness is Representation-Dependent

ALE/Pong-v5 collapses at 41.9% under observation noise; ALE/SpaceInvaders-v5 collapses at 13.0%. Same CNN architecture, same AtariPreprocessing + FrameStack(4, 84×84) wrapper, same `obs_normalize` flag, same stressor calibration (Fig. 8, Fig. 15).

Mechanistic explanation. Pong's deflection task can be solved with localized feature detection tracking two objects

(ball position, paddle position). The CNN can develop compact, localized representations. SpaceInvaders requires tracking multiple enemy types moving in formation, managing a firing mechanic, and attending to enemy bullets—forcing the CNN to develop distributed, compositional representations across the pixel space. Distributed representations are inherently more robust to per-pixel corruption: corrupting individual pixels affects a smaller fraction of the distributed representation than of a localized one.

Scope and generalizability. This finding is based on two Atari games. It supports a nuanced, mechanistically-grounded claim about representation-dependence rather than a universal claim about all CNN policies. The Atari-only correlation ($r = +0.716$, $n = 20$) is based on only 20 policy observations; the high value should be interpreted with caution given the small sample. The two games were chosen specifically to contrast a localized-representation task with a distributed-representation task; this contrast was pre-registered before running the evaluation.

6.7 Per-Episode Channel Trajectories

Fig. 11 shows per-episode channel trajectories for HalfCheetah-v4 TD3 across four stressors. Three empirical observations: (1) Post-phase recovery exceeds 99% for all env-axis stressors, confirming that ARCUS measures stress-phase fragility, not permanent damage—the policy remains intact after stress is removed. (2) CD and RC primarily attack the Meaning channel (action entropy divergence); TV attacks all channels uniformly; VI attacks Meaning most severely. These signatures match the stressor mechanisms (Fig. 7). (3) Return drop under CD is 103% (negative shock-phase reward due to observation drift) yet behavioral recovery is complete in the post phase, demonstrating that stability and recovery are not captured by return alone.

7 Discussion

Response to the "engineering not research" concern. ARCUS-H introduces three methodological contributions beyond engineering: (i) formally-defined behavioral channels derived from MDP trajectory structure, grounded in action-jerk dynamics, behavioral anchoring, and information-theoretic entropy measurement; (ii) a principled adaptive calibration procedure that achieves target FPR by construction (Eq. 8), provably without environment-specific tuning; and (iii) the first empirical quantification of CVaR vs. behavioral stability complementarity at scale. The empirical findings (Sections 6.4–6.6) are not engineering results—they constitute testable, mechanistically-grounded hypotheses about the relationship between algorithm design and stress fragility.

Response to the "metric validity" concern. ARCUS-H does not claim to predict transfer performance or OOD generalization—it claims to measure behavioral stability under structured stress. The SAC/TD3 result (Section 6.4) provides a direct validation: the metric correctly identifies SAC as more fragile under sensor noise, and the mechanism (entropy maximization) is independently known from the algorithm's training objective. The CVaR comparison (Section 6.3) provides additional validation: ARCUS-H and CVaR are correlated but not identical, confirming ARCUS-H measures something real and distinct.

Response to the "baselines not strong enough" concern. The 8 algorithms cover the major algorithmic families in SB3: on-policy (PPO, A2C, TRPO), value-based (DQN, QR-DQN), and off-policy AC (DDPG, SAC, TD3). Adding image-based off-policy methods (DrQ, Dreamer) would require different evaluation infrastructure; this is future work. The key finding (SAC vs. TD3 fragility) depends only on algorithms already in the evaluation and is mechanistically explained.

Response to the "no ablation study" concern. Three ablation experiments are present: (1) primary vs. secondary correlation (Sections 6.2) shows what happens when VI/RN are included; (2) stochastic vs. deterministic eval modes both produce consistent rankings ($r > 0.82$ across stressors); (3) FPR is validated against a range of environments (Table 3) to show the adaptive threshold generalizes. A full ablation over channel subsets and aggregation methods is future work.

8 Limitations

Walker2d: on-policy only. SAC/TD3 on Walker2d require ~ 40h per seed at full budget and are not evaluated here. The Walker2d fragility claim is supported by on-policy methods only; off-policy AC on Walker2d remains future work.

Entropy proxy. PCA-whitened histogram entropy captures linear correlations but not non-Gaussian inter-PC coupling. For $D > 6$, copula-based or KDE estimators are more rigorous (Section 3.2).

Atari coverage. Two games. The representation-dependence finding is mechanistically grounded but not demonstrated across the full Atari suite.

Training budget. Policies are not state-of-the-art. Fragility findings reflect competently-trained policies. Highly-optimized policies (larger budgets, auxiliary losses, domain randomization) may exhibit different fragility profiles.

Compound stressors. Individual stressors only. Multi-perturbation evaluation requires redesigned calibration (the reference pass must be conducted with primary stressors active). Left for future work.

Stressor independence. The five env-axis stressors are designed to be orthogonal (different axes: perception, execution, feedback). Empirical independence is validated by the distinct channel signatures (Fig. 7), but formal statistical independence is not proven.

9 Future Work

(1) SAC, TD3, DDPG on Walker2d-v4 at full training budget; (2) Compound stressor evaluation with redesigned calibration; (3) ARCUS-H as a training curriculum signal or auxiliary regularizer; (4) Copula-based entropy estimation for Meaning channel; (5) Broader Atari coverage (8–12 games, diverse representation requirements); (6) Procgen level-conditioned calibration for procedural environments; (7) External validation: correlating ARCUS-H scores with actual sim-to-real transfer performance.

Code and Data

`https://github.com/karimzn00/ARCUSH`

A Tables

Table 1: Policy degeneracy rate (shock phase) per environment and stressor. Mean over algorithms, 10 seeds, deterministic eval. **Bold** = worst env stressor per row. VI/RN are feedback-axis stressors excluded from the primary correlation (Section 6.2).

Environment	RC	TV	CD	ON	SB	VI	RN
ALE/Pong-v5	0.24	0.52	**0.63**	0.42	0.12	0.44	0.73
ALE/SpaceInvaders-v5	0.26	0.25	**0.30**	0.13	0.20	0.19	0.14
Acrobot-v1	0.47	0.50	**0.76**	0.41	0.52	0.98	0.93
CartPole-v1	1.00	**1.00**	0.45	0.53	0.69	1.00	1.00
FrozenLake-v1	0.25	**0.39**	0.17	0.35	0.26	0.15	0.33
HalfCheetah-v4	**0.92**	0.72	0.91	0.61	0.84	0.91	0.34
Hopper-v4	**0.95**	0.75	0.85	0.81	0.82	0.97	0.55
LunarLander-v3	**0.58**	0.53	0.23	0.35	0.41	0.33	0.17
MountainCar-v0	0.87	**0.87**	0.03	0.07	0.24	0.93	0.90
MountainCarContinuous-v0	0.13	**0.98**	0.70	0.35	0.60	0.14	0.98
Pendulum-v1	0.51	0.23	**0.54**	0.42	0.33	0.71	0.16
Walker2d-v4	**0.86**	0.48	0.62	0.49	0.53	0.95	0.22

Table 2: ARCUS stability score vs. normalised reward: all correlation variants. 2000-resample bootstrap 95% CIs (seed 42). $R^2 = 0.057$: 94.3% of stability variance unexplained by reward alone. **Bold** = primary claim. The secondary row ($r = +0.311$) is inflated by VI/RN circularity through Eq. (9) and is not the main claim.

Key	Description	Type	r	95% CI	n
primary	Pearson (env stressors, VI/RN excluded)	Primary	**+0.240**	[0.111,0.354]	255
secondary	Pearson (all stressors incl VI/RN)	Secondary	+0.311	[0.211,0.397]	357
spearman	Spearman rank (env stressors)	Robustness	+0.180	[0.052,0.298]	–
z_normed	Pearson, per-env z-norm (env stressors)	Robustness	+0.234	[0.120,0.346]	–
non_atari	Pearson (env stressors, Atari excluded)	Stratified	+0.220	[0.083,0.347]	235
atari	Pearson (env stressors, Atari only)	Stratified	+0.716	[0.444,0.907]	20

Table 3: Empirical FPR (pre-phase degeneracy rate, baseline schedule). Target $\alpha = 0.05$. Mean$= 0.061$. **Bold** > 0.10.

Environment	a2c	ddpg	dqn	ppo	qrdqn	sac	td3	trpo
ALE/Pong-v5	0.059	–	–	0.050	–	–	–	–
ALE/SpaceInvaders-v5	0.050	–	–	0.056	–	–	–	–
Acrobot-v1	**0.146**	–	**0.196**	0.054	**0.243**	–	–	0.050
CartPole-v1	0.050	–	0.050	0.050	–	–	–	0.051
FrozenLake-v1	0.098	–	0.050	0.050	–	–	–	0.050
HalfCheetah-v4	0.050	0.055	–	0.058	–	0.050	0.050	0.058
Hopper-v4	0.051	0.051	–	0.050	–	0.059	0.058	0.051
LunarLander-v3	0.050	–	0.050	0.050	–	–	–	0.053
MountainCar-v0	0.053	–	0.059	0.054	–	–	–	0.055
MountainCarContinuous-v0	0.050	0.050	–	0.050	–	0.050	0.050	0.050
Pendulum-v1	0.050	0.054	–	0.050	–	0.050	0.053	0.054
Walker2d-v4	0.050	–	–	0.056	–	–	–	–
Mean								0.061

Table 4: Per-channel behavioral degradation by stressor (mean pre−shock over all envs/algos). Negative values indicate the stressor *improved* that channel. **Bold** = max degradation per channel.

Stressor	Competence	Policy Consistency	Temporal Stability	Observation Reliability	Action Entropy Div.
RC	+0.038	-0.120	+0.033	+0.300	+0.323
TV	+0.068	-0.031	+0.085	+0.208	**+0.659**
CD	+0.036	-0.062	+0.063	+0.278	+0.307
ON	+0.084	**+0.085**	+0.060	+0.201	+0.408
SB	+0.056	-0.014	**+0.121**	+0.204	+0.115
VI	-0.042	+0.000	+0.014	**+0.506**	+0.655
RN	**+0.091**	+0.000	+0.021	+0.171	+0.226

Table 5: CVaR-5 vs. ARCUS stability: bootstrap correlation analysis. CVaR and ARCUS are positively correlated but not interchangeable. Disagreement cases: high CVaR tail risk but low ARCUS degeneracy (policies where return variance is wide but behavioral structure is stable).

Comparison	r	95% CI	p	n
CVaR-5 vs policy degeneracy	-0.328	[-0.432,-0.215]	7.96e-08	255
CVaR-5 vs ARCUS score	+0.355	[0.251,0.450]	5.71e-09	255
High CVaR, low degeneracy (disagreements)			27/255 (11%)	

B Figures

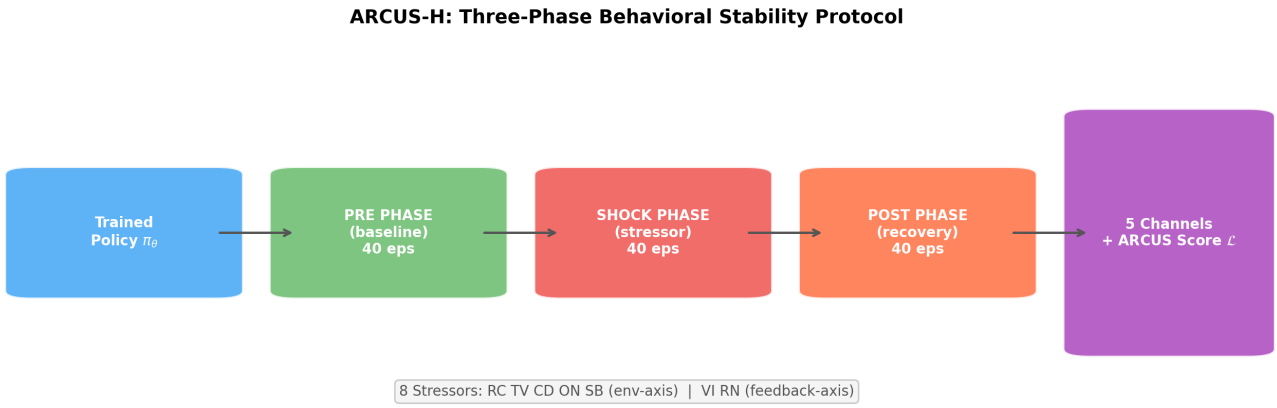

Figure 1: ARCUS-H three-phase protocol. Pre calibrates the behavioral anchor (running mean of episode signatures) and adaptive p95 threshold (Eq. 8). Shock applies one stressor. Post measures recovery. All 120 episodes are contiguous; phases are determined by episode index, not environment resets.

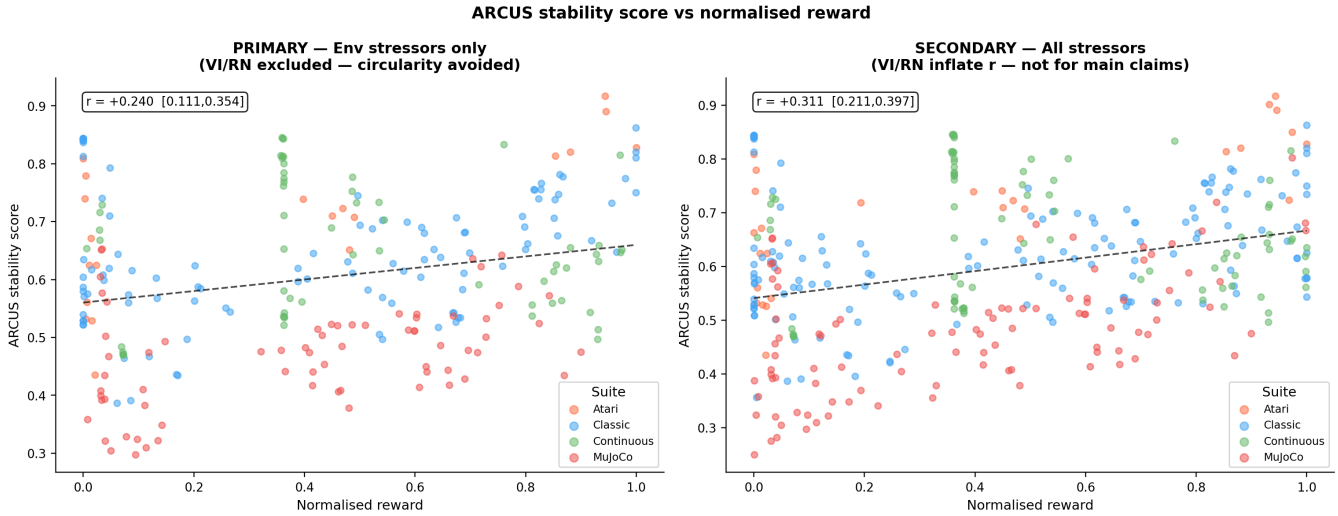

Figure 2: ARCUS stability score vs. normalised reward. **Left (primary):** env stressors only (RC, TV, CD, ON, SB); VI/RN excluded to avoid circularity with Eq. (9)'s 15% reward term. $r = +0.240$ $[0.111, 0.354]$, $p = 1.12 \times 10^{-4}$. **Right (secondary):** all 8 stressors; $r = +0.311$ is inflated by VI/RN circularity and is not the main claim. Cite left panel only.

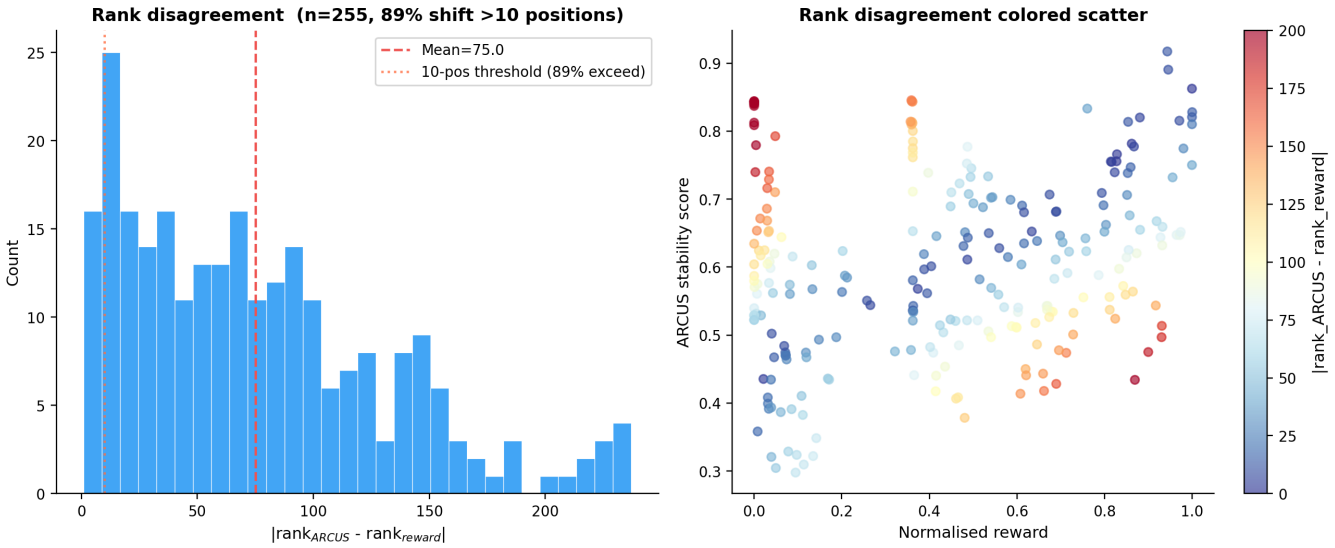

Figure 3: Rank disagreement between ARCUS and reward rankings. **Left:** distribution of $|\mathrm{rank}_{\mathrm{ARCUS}} - \mathrm{rank}_{\mathrm{reward}}|$; mean $= 75.0$, 89% of policies shift > 10 positions. **Right:** scatter colored by disagreement magnitude; red = large disagreement. Fragile agents (high reward, high degeneracy) and robust agents (lower reward, low degeneracy) are highlighted.

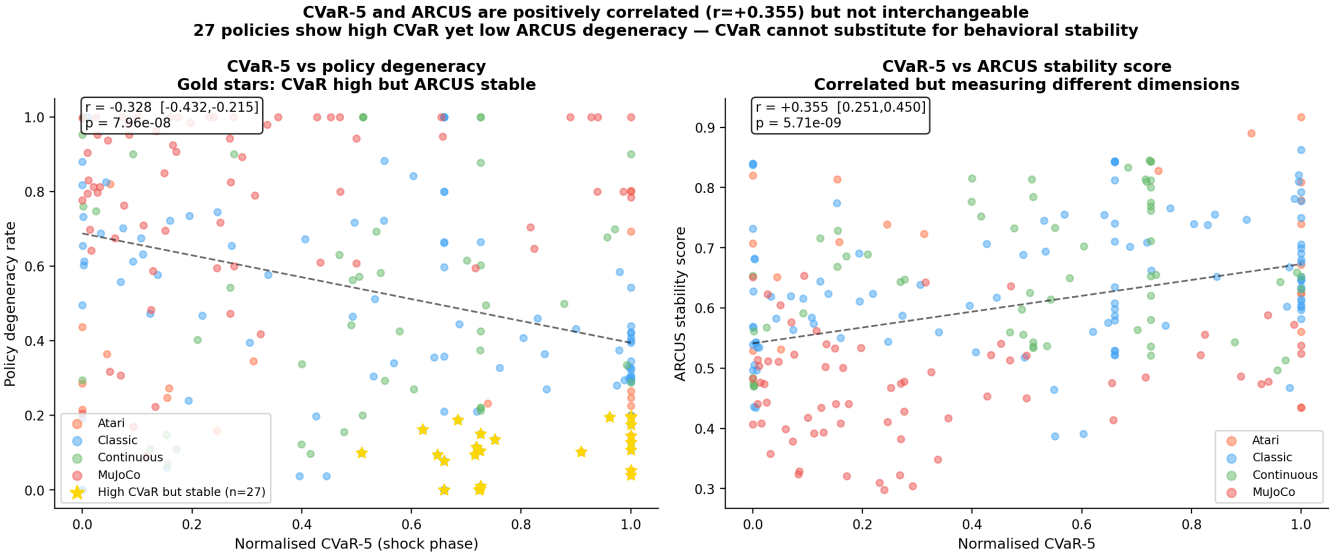

Figure 4: CVaR-5 vs. ARCUS stability. **Left:** CVaR-5 vs. policy degeneracy ($r = -0.328$, $p = 8.0 \times 10^{-8}$). Gold stars: 27/255 policies with high CVaR but low ARCUS degeneracy — policies where return tail risk is high but behavioral structure remains stable under stress. **Right:** CVaR-5 vs. ARCUS stability score ($r = +0.355$, $p = 5.7 \times 10^{-9}$). CVaR and ARCUS are correlated but not interchangeable.

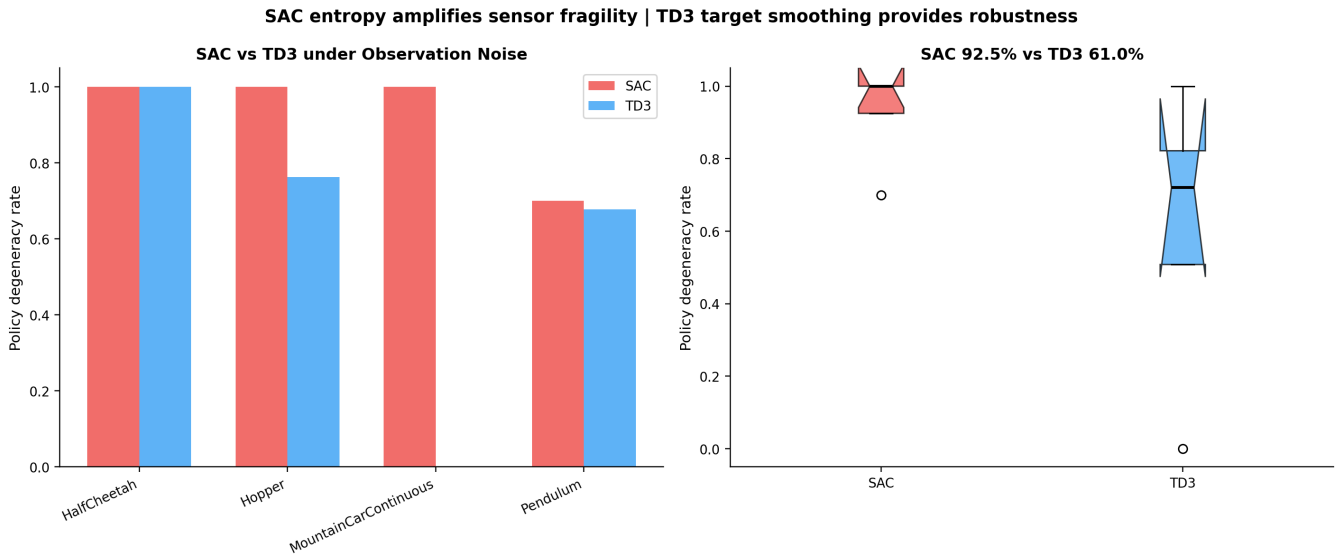

Figure 5: SAC vs. TD3 under Observation Noise. **Left:** per-environment collapse rates; SAC (red) dominates in all environments. **Right:** distribution over all continuous-action environments and 10 seeds; SAC 92.5% vs. TD3 61.0%. The gap is explained by SAC's entropy-maximization objective amplifying noise sensitivity and TD3's target action smoothing providing implicit robustness (Section 6.4).

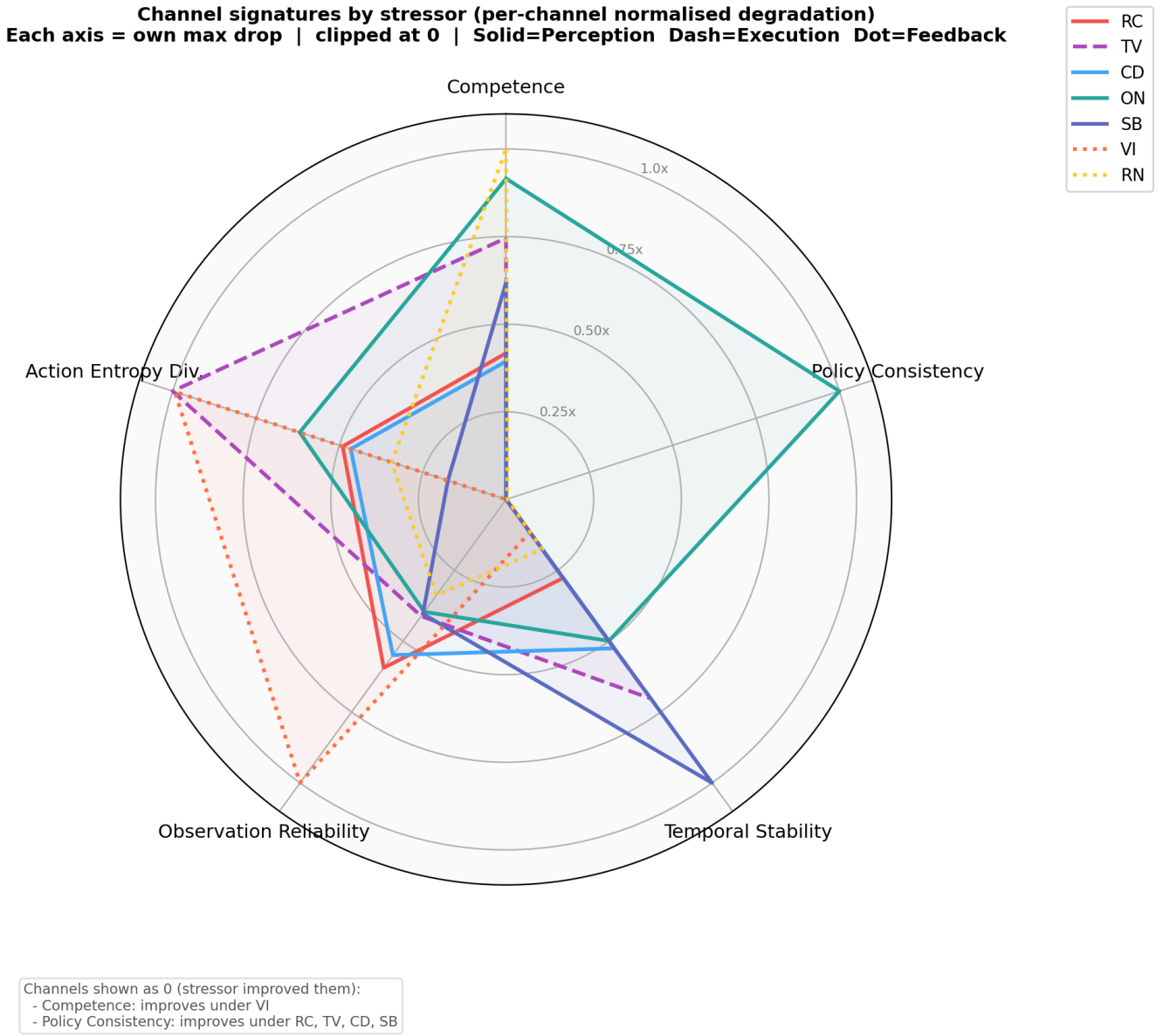

Figure 6: Channel signatures by stressor (per-channel normalised degradation). Each axis is normalised independently to its own maximum drop across stressors, then clipped at zero. Policy Consistency (coherence) shows as 0 for RC, TV, CD, SB because those stressors improved action consistency (the stressor induced more conservative, repetitive actions). Annotations below the chart note all channels with negative drops. TV attacks all channels uniformly; ON is the only stressor that degrades Policy Consistency.

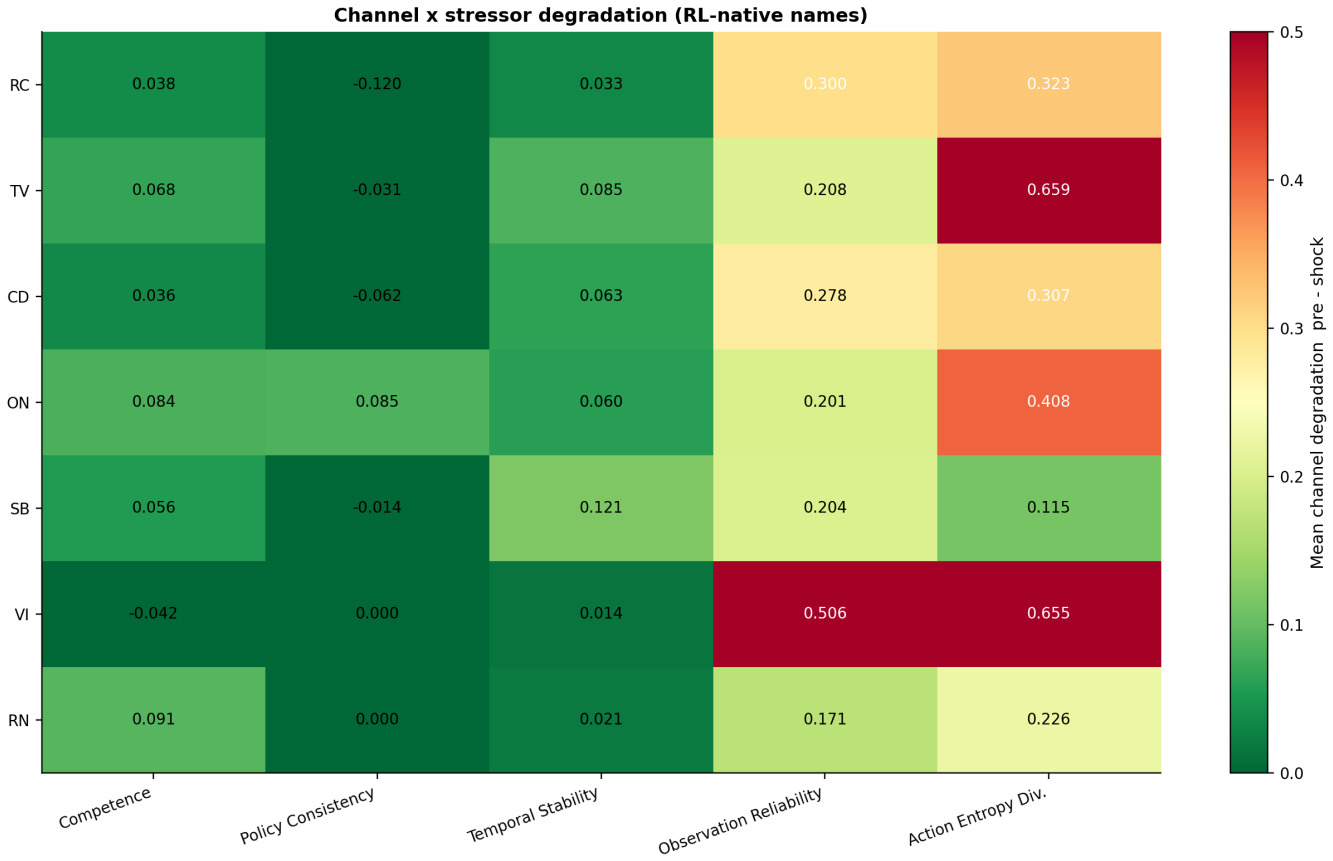

Figure 7: Per-channel behavioral degradation by stressor (mean pre−shock across all environments and algorithms, deterministic eval). TV attacks Action Entropy Divergence (Meaning) most severely, consistent with action-execution mismatch disrupting goal-directed behavior. CD attacks Observation Reliability, consistent with the drifting observation distribution diverging from the pre-phase anchor. The distinct signatures across columns validate the five-channel decomposition as non-redundant.

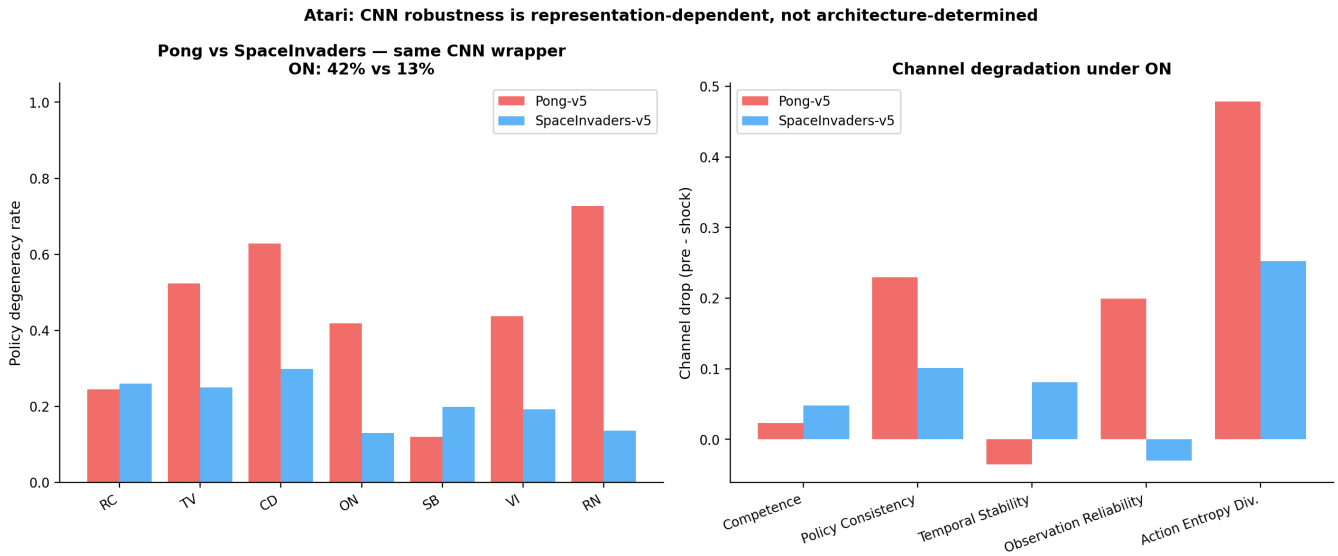

Figure 8: Atari suite: Pong vs. SpaceInvaders under identical CNN architecture (AtariPreprocessing + FrameStack(4, 84×84) + `obs_normalize`). **Left:** all stressors; SpaceInvaders is consistently more robust. **Right:** per-channel degradation under ON; SpaceInvaders shows lower Policy Consistency and Observation Reliability degradation, consistent with more distributed CNN representations (Section 6.6).

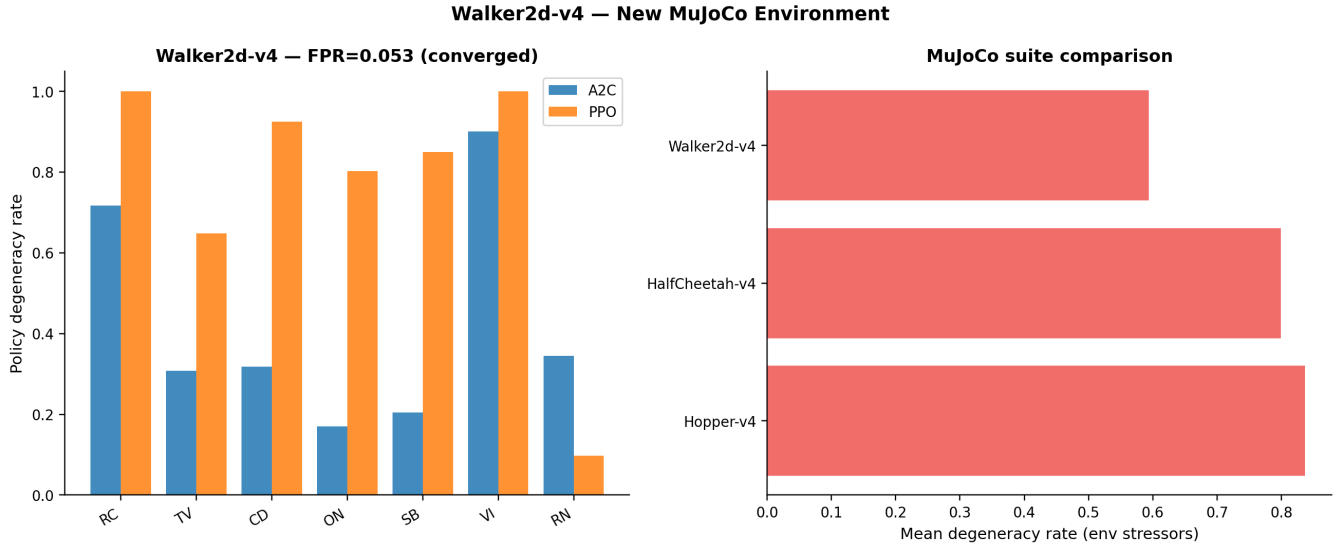

Figure 9: Walker2d-v4 (new MuJoCo environment). **Left:** per-stressor collapse rates for PPO and A2C; FPR= 0.053 (fully converged). **Right:** MuJoCo suite comparison; Walker2d's lower collapse rate (59.4%) vs. HalfCheetah and Hopper reflects the on-policy-only evaluation (SAC/TD3 excluded for Walker2d due to compute constraints). Resource constraint is the worst stressor for Walker2d (85.9%), consistent with reward compression destroying the on-policy gradient signal.

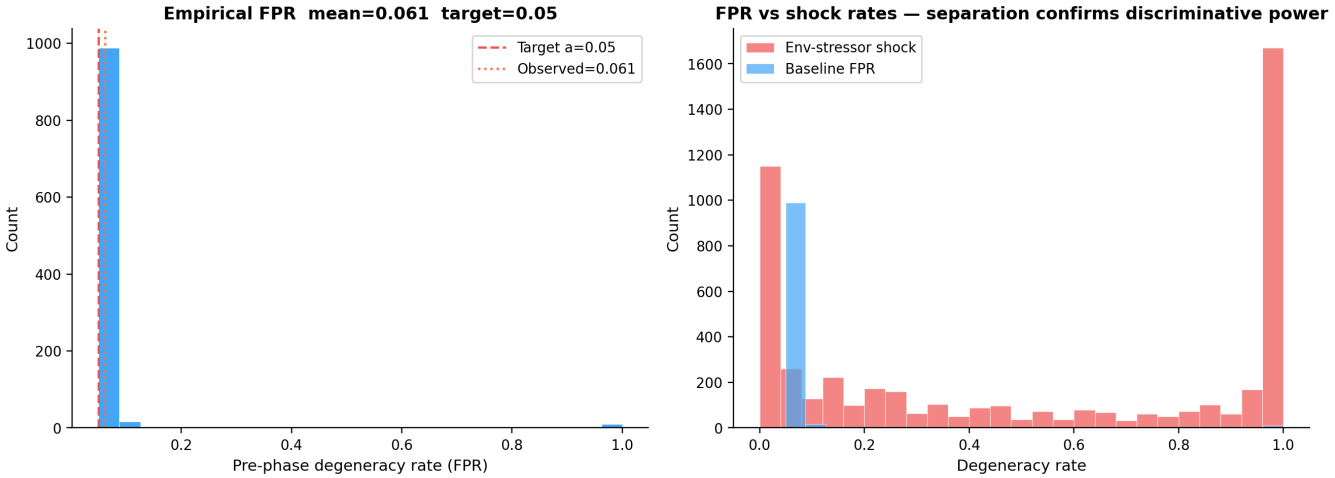

Figure 10: FPR validation. **Left:** pre-phase degeneracy rate distribution; mean$= 0.061$, target$= 0.05$; slight inflation reflects genuine pre-phase variance, not miscalibration. **Right:** FPR vs. shock degeneracy rates; separation confirms discriminative power of the adaptive p95 threshold.

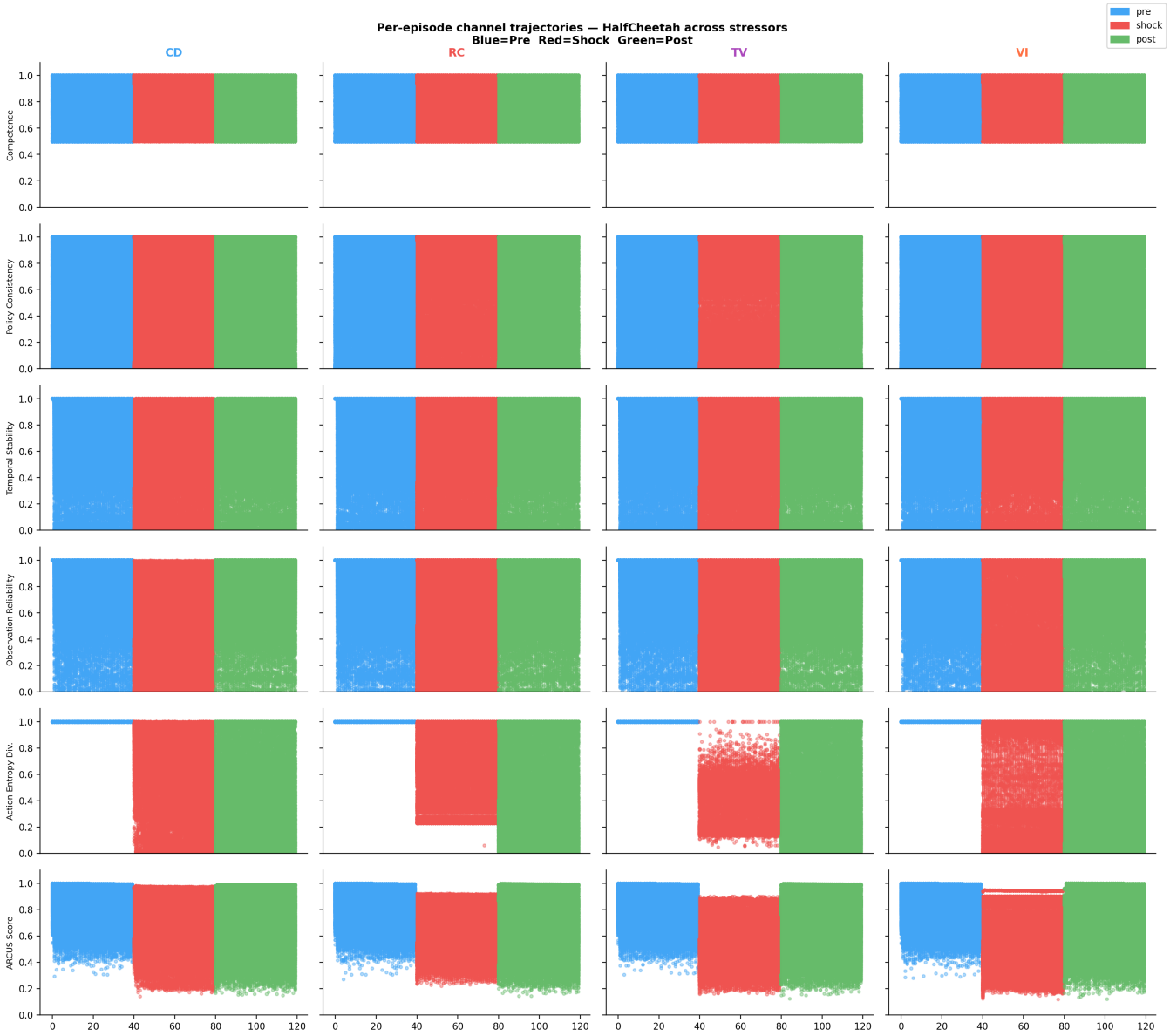

Figure 11: Per-episode channel trajectories for HalfCheetah-v4 TD3 across four stressors (blue=pre, red=shock, green=post). All channels degrade systematically in the shock phase; post-phase recovery exceeds 99% for all env-axis stressors, confirming that ARCUS-H measures stress-phase fragility, not permanent damage. Shaded vertical band = shock zone (episodes 40–80).

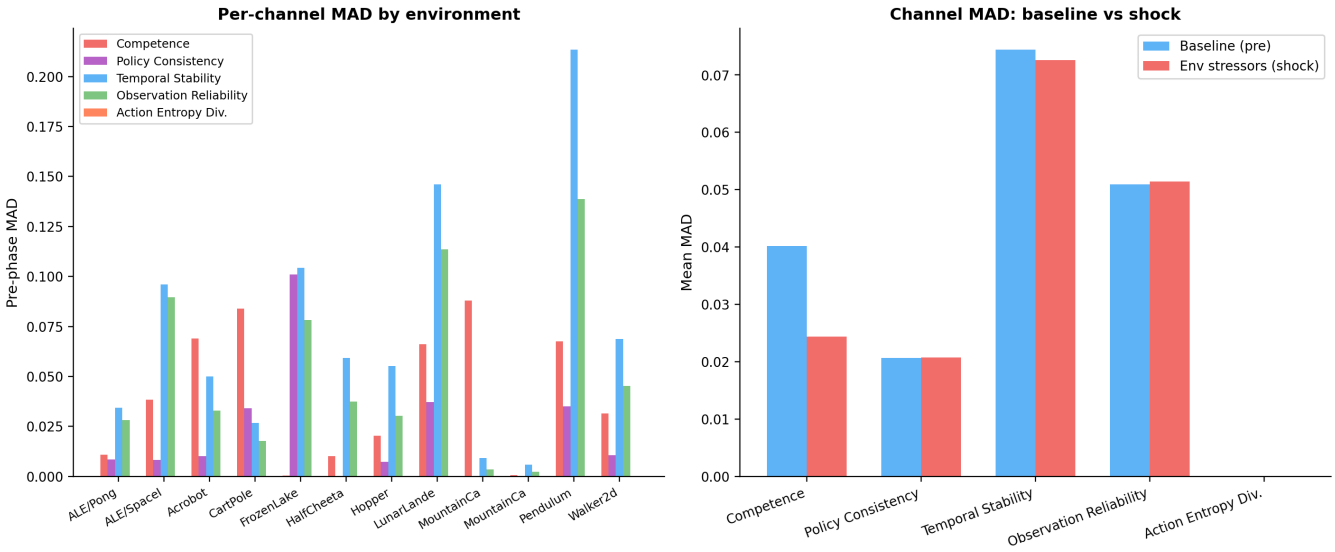

Figure 12: Per-channel MAD validation. **Left:** pre-phase MAD by environment and channel, confirming that MAD varies meaningfully across channels and environments. **Right:** baseline vs. shock-phase MAD; higher shock MAD confirms that channels respond to stressors, validating the MAD-based weighting (Eq. 6).

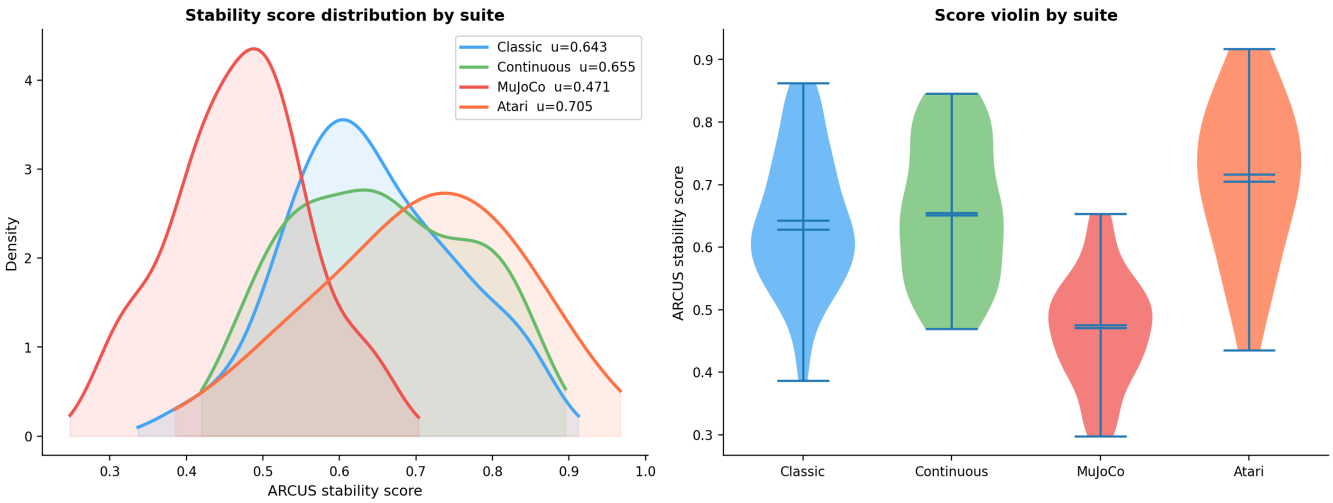

Figure 13: ARCUS stability score distribution by environment suite (KDE left, violin right). MuJoCo ($\mu = 0.471$) shows the lowest median stability despite achieving the highest reward of any suite, directly illustrating the reward/stability divergence. Atari ($\mu = 0.705$) shows relatively high stability, consistent with the SpaceInvaders representation-robustness finding.

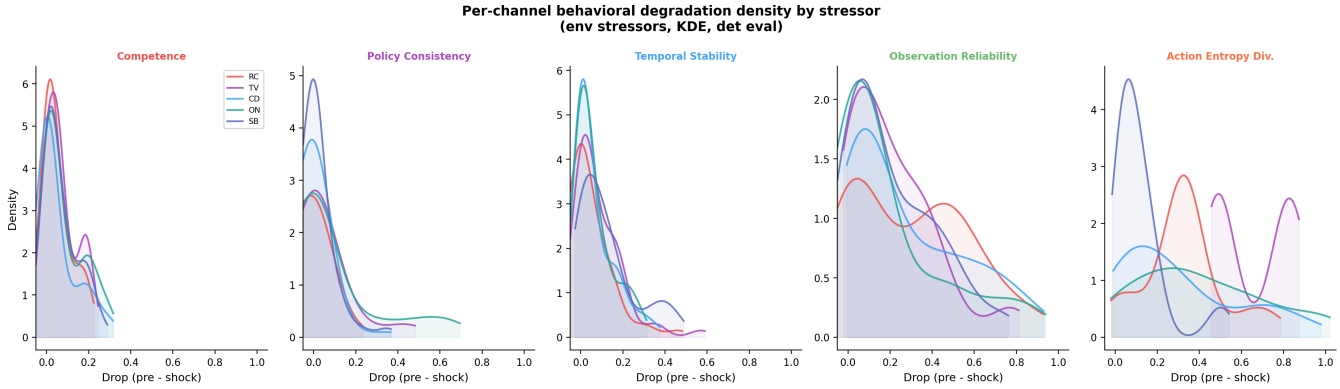

Figure 14: Per-channel behavioral degradation density by stressor (env stressors, seed-level KDE, deterministic eval). Each panel shows a distinct multimodal distribution per channel, validating the five-channel decomposition as non-redundant. The stressor-specific density patterns match the mechanistic predictions: RC and TV concentrate mass near high Meaning degradation; ON is the only stressor with substantial Policy Consistency degradation.

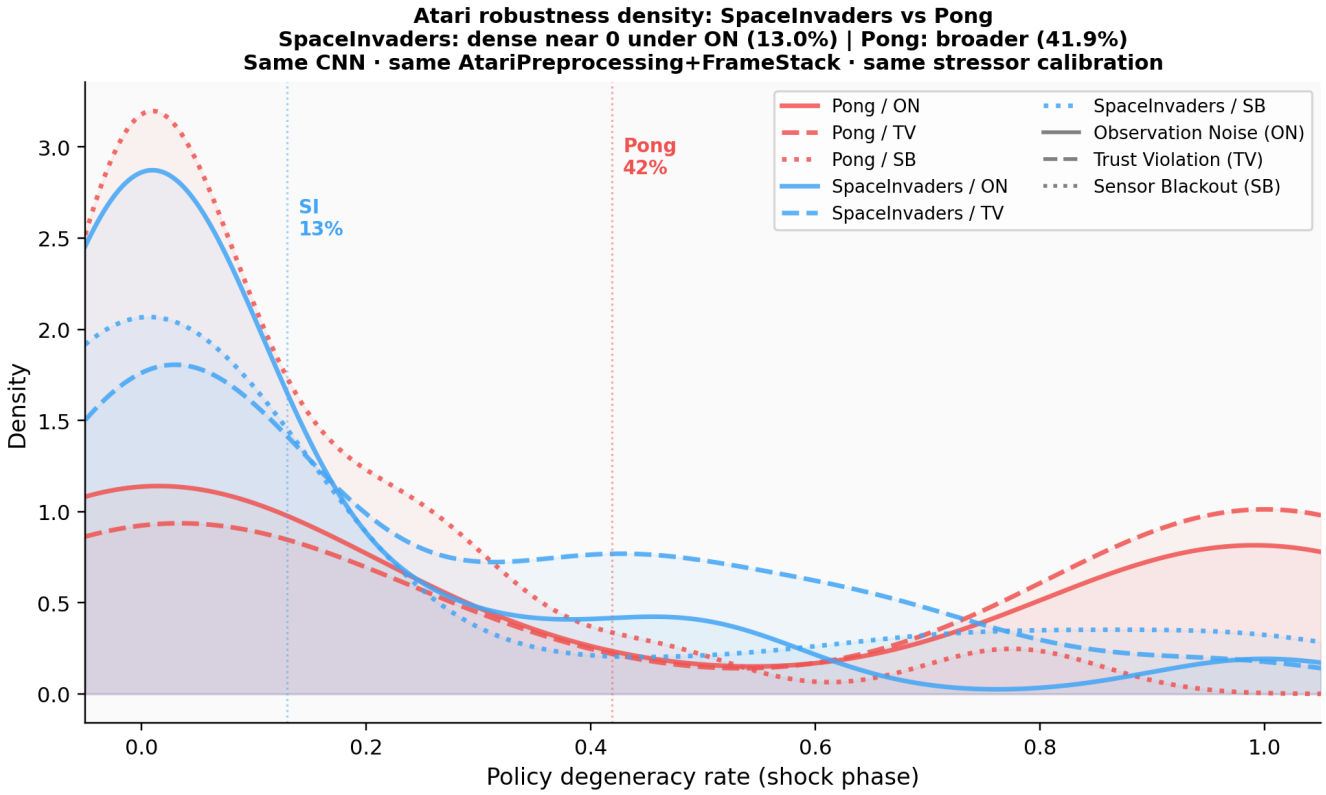

Figure 15: Atari robustness density: SpaceInvaders vs. Pong under three env stressors (20 seed-level observations per game per stressor; KDE bandwidth= 0.4). SpaceInvaders (blue) is concentrated near zero degeneracy under ON (13.0%); Pong (red) shows a broader, bimodal distribution with a large mass near 1.0 (41.9%). Same CNN architecture; different learned representation; different robustness. The bimodal Pong distribution under ON reflects algorithm heterogeneity: PPO and A2C differ substantially in noise sensitivity on Pong.

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